

Project Proposal: IR-Guided Docking Station for a Mobile Robot

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1. Team Information

Field	Value
Team name	Group 1
Team members	Santeri Karjalainen, Zohaib Salman
Team leader	Santeri Karjalainen
Initial project area	Autonomous docking for a mobile robot (embedded systems / robotics)

2. Problem Statement

A robot cannot charge itself: its batteries must be removed and charged by hand. Its availability therefore depends on someone remembering to handle charging, and the robot is out of service whenever its batteries are depleted or being swapped.

The problem occurs wherever the robot operates, including during development and testing. It affects the people who operate and maintain the robot, and anyone who relies on the robot while it is out of service.

It matters because a robot that cannot keep itself operational adds maintenance work and cannot run unattended for long periods.

3. Users

Who benefits:

Stakeholder	Benefit
Robot operators	No battery handling or charging routines to remember
End users of the robot's services	Higher robot availability, fewer service interruptions
Maintenance / technical staff	Less connector wear and battery handling. Predictable parking location
Robot developers	Long unattended test runs; docking subsystem reusable on other platforms

4. Current Situation

Charging is fully manual today. The robot uses Li-ion packs that are removed and charged in an external charger. A person must notice the low battery, access and disconnect the packs, charge them, and reinstall them.

Difficulties and limitations:

- Charging depends on human attention.
- Manual battery handling takes operator time and causes connector wear and handling risk.
- The robot has no defined home position.
- Battery life depreciates if left unattended.

5. Proposed Improvement

A docking station the robot finds and connects on its own. When the battery is low or the robot is idle, it returns to the station, aligns itself, and makes electrical contact without human help.

What improves:

- The robot keeps itself operational without human involvement.
- The robot has a fixed, predictable parking location.
- The robot overall life span increases.

6. Expected Value

Autonomous docking turns the robot from a device people must maintain into one that maintains itself.

- **Reduced manual work:** eliminates routine battery handling.
- **Improved availability:** the robot is charged and ready more of the time, including after overnight idle periods.
- **Improved battery life:** the battery is charged up to a certain threshold and is kept at that percentage like a laptops optimized battery charging system.
- **Improved safety:** less manual handling of Li-ion packs and connectors; the robot parks out of walkways instead of stopping wherever its battery runs out.
- **Product readiness:** autonomous docking is a baseline expectation for mobile service robots and a prerequisite for long unattended operation.

7. Initial Technical Challenge

The core challenge is precise final alignment. General navigation can bring the robot near the station, but mating charging contacts needs centimetre-level position and a few degrees of heading accuracy.

The project combines embedded firmware, signal encoding and filtering against ambient IR, closed-loop motion control, mechanical design of self-aligning contacts, and integration with the robot's existing motor control and power systems.

8. Project Scope

The MVP proves the robot can dock itself from a defined starting area and make electrical contact.

In scope:

- A docking station with an IR beacon emitting distinguishable signals (left / centre / right zones).
- IR receivers on the robot and firmware that steers it onto the station's centre line.
- A docking behaviour starting within a defined area in front of the station and ending parked on the contacts.
- Contact detection confirming a successful dock.

Success criterion: the robot docks in at least [8 of 10] trials from varied starting positions within the defined area, under normal indoor lighting.

Stretch goal: charging the batteries through the dock contacts. This requires an onboard 3S charge path (balancing, current limiting, BMS interaction) and is kept out of the MVP.

Out of scope: finding the station from anywhere in the building (map-based navigation), obstacle avoidance during docking, and automatic undocking or scheduling logic.